

CS 217: Artificial Intelligence and Machine Learning (Quiz)

Feb 6, 2026.
8:30 – 9:30am

Roll: _____
Name: _____
Mode: Credit/Audit/Sit-through _____

Write all your answers in the space provided. Do not spend time/space giving irrelevant details or details not asked for. Use the marks as a guideline for the amount of time you should spend on a question. You are only allowed to refer your hand-written notes, no one else's notes or textbook.

1. Consider training a linear regression model with general L_γ regularizer defined as $\sum_{j=1}^d |w_j|^\gamma$ for $\gamma \geq 0$. In class we considered $\gamma = 2$ calling it Ridge regularizer and $\gamma = 1$ calling it Lasso regularizer. Is the training objective for linear regression with L_γ regularizer convex in $\mathbf{w}$ for all $\gamma \geq 0$? ..2

The objective is convex only for $\gamma \geq 1$ and non-convex for $0 \leq \gamma < 1$.

Updated Rubric (Total: 2 marks):

- Correctly identifying the convexity threshold (convex for only $\gamma \geq 1$ or non-convex for $\gamma < 1$): **1 marks**. (Correctly identifying the threshold $\gamma = 1$ demonstrates understanding of where curvature changes. It is unlikely that a student can determine the correct range without reasoning about the second derivative or convexity properties, even if the proof is not explicitly written. In such cases awarded some marks.)
- Providing any valid justification for either case (e.g., second derivative argument, Hessian sign argument, violation of convexity inequality, counterexample, or curvature reasoning): **1 mark**.

Important Clarifications:

- It is NOT necessary to explicitly prove both directions. Correct identification of the threshold $\gamma = 1$ is sufficient for full conceptual credit also. And proof just showing violation without range in correct manner, is also considered.
- If the final classification is correct but reasoning is vague or partially incorrect, awarded at most **1 mark**.
- If a student provides a valid counterexample (e.g., midpoint inequality violation for $\gamma < 1$), full marks may be awarded even without formal proof.
- Writing only “Yes/No” or “Convex/Not convex” without discussing the role of γ receives **0 mark**.

Common Conceptual Traps Observed:

(a) **Parity (Odd/Even) Argument**

Claiming convexity depends on whether γ is odd or even (e.g., “even powers are convex, odd powers are not”).

This is incorrect because the regularizer is $|w|^\gamma$, not w^γ . Convexity depends on whether $\gamma \geq 1$, not on parity.

Marks: 0

(b) **Hessian Must Be Constant**

Claiming that since the Hessian depends on w , the function is not convex.

Convexity does NOT require the Hessian to be constant. It only requires the Hessian to be positive semidefinite for all w .

Marks: 0 or at most 1 (if classification correct).

(c) **“Hessian Depends on w Hence Not PSD”**

Arguing that because the second derivative or Hessian depends on w , it may not remain positive.

Dependence on w does not invalidate convexity. The sign of the Hessian must be checked explicitly. For $\gamma \geq 1$, it remains non-negative everywhere.

Marks: 0 unless final classification is correct (then at most 1).

(d) **Using w^3 Instead of $|w|^3$**

Computing derivatives of w^3 instead of $|w|^3$ and concluding non-convexity.

This is a fundamental misunderstanding of the regularizer.

Marks: 0

(e) **No Case Distinction**

Writing “Not convex” or “Convex” without specifying the role of γ .

Marks: 0

(f) **Correct Counterexample for $\gamma < 1$**

Providing a valid midpoint violation or curvature argument demonstrating non-convexity for $\gamma < 1$.

Full marks.

Answer: Convex only for $\gamma \geq 1$. Show proof that for $\gamma < 1$ objective is non-convex, and for $\gamma \geq 1$ it is convex.

2. Provide an estimate of the number of iterations needed to find the optimal $\mathbf{w}$ when training the ordinary linear regression model using Newton’s second order method using the full data, that is using the following optimization algorithm:

```
t = 0,  $\mathbf{w}^t = 0$ 
Let  $F(\mathbf{w}) = \sum_i (y_i - \mathbf{w} \cdot \mathbf{x}^i)^2$ 
while  $\|\nabla F(\mathbf{w}^t)\| \neq 0$  do
     $\mathbf{w}^{t+1} = \mathbf{w}^t - [H_F(\mathbf{w}^t)]^{-1} \nabla F(\mathbf{w}^t)$ 
    t = t + 1
end while
```

..2 One iteration suffices.

Rubric: Full mark for mentioning number of iteration. Partial mark for only derivation

without answering (or if it is not clear) number of iterations.

Derivation:

$$H_F(w^t) = \nabla_w^2 L(w) = \nabla_w^2 \|xw - y\|^2 = \dots = 2x^T x;$$

$$\nabla F(w^t) = 2x^T (xw^t - y)$$

$$Pass1 : w^1 = w^0 - (x^T x)^{-1} x^T (xw^0 - y) = (x^T x)^{-1} x^T y \quad \text{which is a OLS objective}$$

$$\|\nabla F(w^t)\| = 0 \quad \text{Hence, one iteration only.}$$

3. Suppose you have an initial training dataset $D = (X, Y)$ where $X \in \mathbb{R}^{N \times d}, Y \in \mathbb{R}^N$ using which you train a linear regression model using Bayesian estimation with prior probability on weights defined as $p_0(\mathbf{w}) = \mathcal{N}(\mathbf{w}; 0, \eta I)$. Now you get some more examples D' . Instead of training the parameters from scratch using the union of D and D' , you decide to use the posterior distribution $P(\mathbf{w}|D)$ as a prior distribution $p_1(\mathbf{w})$ for estimating $P(\mathbf{w}|D', p_1)$ using Bayesian estimation. Assume the true label y given an $\mathbf{x}$ follows a Gaussian distribution where $P(y|\mathbf{w}) = \mathcal{N}(y; \mathbf{w}^T \mathbf{x}, \sigma^2)$. In answering the questions below you may find the following properties of Gaussian distributions useful.

If $P(A) = \mathcal{N}(A; \mu_A, \Sigma_A), P(B|A) = \mathcal{N}(B; MA, \Sigma_B)$ then

$$P(A|B) = \mathcal{N}(A; \mu_{A|B}, \Sigma_{A|B}), \text{ where } \Sigma_{A|B} = (\Sigma_A^{-1} + M^T \Sigma_B^{-1} M)^{-1}, \mu_{A|B} = \Sigma_{A|B} (M^T \Sigma_B^{-1} B + \Sigma_A^{-1} \mu_A)$$

- (a) Show that the posterior $P(\mathbf{w}|D)$ follows a Gaussian distribution ..2 Since the likelihood is Gaussian ($P(Y|X, \mathbf{w}) = \mathcal{N}(Y; X\mathbf{w}, \sigma^2 I)$) and the prior is Gaussian ($p_0(\mathbf{w}) = \mathcal{N}(\mathbf{w}; 0, \eta I)$), using the property of Gaussians above with $A = \mathbf{w}, B = Y, M = X$ we get the results that $P(\mathbf{w}|D)$ is Gaussian.

Rubric: Full marks given for the above justification or mentioning multiplication of two gaussian is gaussian. Partial marks given for partially correct intuitions.

- (b) If $P(\mathbf{w}|D) = \mathcal{N}(\mathbf{w}; \mu_1, \Sigma_1)$ derive expression for the mean μ_1 and covariance Σ_1 ..2 the covariance matrix Σ_1 and the mean μ_1 for $P(\mathbf{w}|D) = \mathcal{N}(\mathbf{w}; \mu_1, \Sigma_1)$ are:

$$\Sigma_1 = \left(\frac{1}{\sigma^2} X^T X + \frac{1}{\eta} I \right)^{-1}$$

$$\mu_1 = \frac{1}{\sigma^2} \Sigma_1 X^T Y$$

Rubric: Full marks given for the above derivation (1 mark for each). No partial marks given for incorrect derivation. For minor mistakes 0.5 deducted.

Derivation: From the question we have $w = A, \mu_1 = \mu_{A|B}, \Sigma_1 = \Sigma_{A|B}, \mu_A = 0,$

$$\Sigma_A = \eta I, M = X, \Sigma_B = \sigma^2, B = y$$

$$\text{now, } \Sigma_{A|B} = (\Sigma_A^{-1} + M^T \Sigma_B^{-1} M)^{-1}, \text{ and } \Sigma_1 = ((\eta I)^{-1} + x^T \frac{1}{\sigma^2} x)^{-1} = \left(\frac{1}{\eta} I + \frac{1}{\sigma^2} x^T x \right)^{-1}$$

$$\mu_1 = \mu_{A|B} = \Sigma_{A|B} (M^T \Sigma_B^{-1} B + \Sigma_A^{-1} \mu_A)^{-1} = \Sigma_1 (x^T \frac{1}{\sigma^2} y + \frac{1}{\eta} I 0)^{-1} = \frac{1}{\sigma^2} \Sigma_1 x^T y$$

- (c) Write the expression for the training objective to calculate the most likely value of the posterior of $\mathbf{w}$ after observing D' and D under the above setting. ..2 To find the most likely value (MAP) after observing $D' = (X', Y')$, we maximize the posterior $P(\mathbf{w}|D', D)$, which is equivalent to minimizing the negative log-posterior using $p_1(\mathbf{w})$ as the prior:

$$J(\mathbf{w}) = \frac{1}{2\sigma^2} \sum_{(\mathbf{x}_i, y_i) \in D'} (y_i - \mathbf{w}^T \mathbf{x}_i)^2 + \frac{1}{2} (\mathbf{w} - \mu_1)^T \Sigma_1^{-1} (\mathbf{w} - \mu_1)$$

Rubric: Full marks given for the above answer. Partial marks given for incomplete derivation / partially correct answers.

Derivation: $D = (x, y); P(w|D) = N(\cdot); P_1(w) = N(\cdot)$ and $D' \propto P(y'|x', w)P_1(w)$

$MAP : \operatorname{argmax}_w P(w|D', D) = \operatorname{argmin}_w [-\log P(y'|x', w) - \log P_1(w)]$

here $P(y'|w', w) = \prod_{i=1}^{N'} N(y; w^T x_i, \sigma^2)$ and $P_1(w) = N(w; \mu_1, \Sigma_1)$

$-\log P(y'|x', w) = \frac{1}{2\sigma^2} \sum_{x,y \in D'} (y_i - w^T x_i)^2$ and $-\log P_1(w) = \frac{1}{2}(w - \mu_1)^T \Sigma_1^{-1} (w - \mu_1)$

- (d) Would you get a different result if instead of following the above two-step procedure you directly trained $\mathbf{w}$ with $D \cup D'$ starting with p_0 as prior? ..2

No, you would not get a different result. Bayesian inference is inherently sequential. The posterior calculated using D as the prior for D' yields the same distribution as performing a single update using the pooled data $D \cup D'$ with the original prior p_0 . This is because:

$$P(\mathbf{w}|D, D') \propto P(D'|\mathbf{w})P(D|\mathbf{w})p_0(\mathbf{w}) = P(D'|\mathbf{w})P(\mathbf{w}|D)$$

Rubric: Full mark for mentioning "NO" along with justifications. No partial marks given for any other justifications.

4. Consider the following loss function $L(\mathbf{x}, y, f) = \frac{1}{1+\exp(-yf(\mathbf{x}))}$ called the Sigmoid loss where $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$.

- (a) In a x-y plot show the loss versus $yf(\mathbf{x})$

You have to simply plot a standard sigmoid curve. ..1

- (b) Is the loss function concave, convex, or neither? Justify briefly.

Neither convex nor concave. Find double differential of $g(v) = \frac{1}{1+\exp(v)}$. This turns out to be $g''(v) = \frac{\exp(v)(1-\exp(v))}{(1+\exp(v))^3}$. This quantity is > 0 for some values of v and < 0 for others. Using the results of Q3 we can show that the loss function is neither concave nor convex in $\mathbf{w}$. ..2

- (c) State one major advantage of the sigmoid loss above the logistic loss.

The maximum loss of any instance is 1 whereas it increases almost linearly for Logistic loss. This implies that for noisy training instances, of the kind in Question 2, Logistic loss will give rise to a bad separator.

Common mistakes:

- i. "Since sigmoid is bound between -1 and +1, it is numerically stable". This is not technically correct since calculations can always be made numerically stable by using approximations, clipping values, or computing exponents in log-space. However, partial marks are still given for identifying the bound.

..2

5. Suppose we wish to train a square error regression function with a $L2$ regularizer. That is the training objective is $\min_{\mathbf{w}} F(\mathbf{w})$ where $F(\mathbf{w}) = \sum_i (y_i - \mathbf{w} \cdot \mathbf{x}^i)^2 + \|\mathbf{w}\|^2$. Suppose, we decide to use an iterative optimization algorithm as follows:

$t = 0, \mathbf{w}^t = 0$

while $\|\nabla F(\mathbf{w}^t)\| \neq 0$ **do**

 Choose a direction $\mathbf{e}^t$.

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 $g_t(\lambda) = F(\mathbf{w}^t + \lambda \mathbf{e}^t) - F(\mathbf{w}^t)$ 
 $\lambda^* = \min_{\lambda} g_t(\lambda)$ 
 $\mathbf{w}^{t+1} = \mathbf{w}^t + \lambda^* \mathbf{e}^t$ 
 $t = t + 1$ 
end while

```

Let the labeled data be as shown in the table below.

x_1	x_2	y
2	8	1
4	7	1
1	5	-1
7	2	-1
8	3	-1

- (a) Suppose you are allowed to add an $N + 1$ th training point, what would you choose so that the algorithm terminates without entering the while loop? ..2

Answer: The algorithm won't enter the loop if $\|\nabla F(\mathbf{w}^0)\| = 0$

$$\|\nabla F(\mathbf{w}^0)\| = \sum_i 2(y_i - \mathbf{w}^{0T} \mathbf{x}^i)(-\mathbf{x}^i) + 2\mathbf{w}^0 = [0 \ 0]^T$$

$$\sum_i y_i \mathbf{x}^i = [0 \ 0]^T$$

$$\sum_i y_i x_1^i = 0 \text{ and } \sum_i y_i x_2^i = 0$$

For $y_6 = 1$,

$$x_1^6 = -\sum_i y_i x_1^i \text{ and } x_2^6 = -\sum_i y_i x_2^i, \text{ } i \text{ from } 1 \text{ to } 5$$

which gives $x_1^6 = 10$ and $x_2^6 = -5$

Rubric:

- Correct final point stated with no supporting steps: **0.5 mark**.
- Correct formulation of the stopping condition ($\|\nabla F(\mathbf{w}^0)\| = 0$ or $\sum_i y_i \mathbf{x}^i = 0$) with correct final answer: **1 mark**.
- Complete and correct derivation, including formulation, computation of required sums, choice of $(\mathbf{x}^{N+1}, y^{N+1})$, and clear statement of the added point: **2 marks (full)**.
- Correct steps but incorrect final numerical answer due to minor algebraic or arithmetic mistakes (e.g., sign error, missing term): **1 mark (grace)**.
- Only formulation with no attempt to compute or specify the added point: **0.5 mark**.
- Selecting a point first and then attempting to justify it (reverse order of expected reasoning): **at most 1 mark**.

- (b) For the N points in the table, suppose at $t = 0$ $\mathbf{e}^t = [1 \ 1]^T$, what is $g_0(\lambda)$. (Make sure $g_0(\lambda)$ contains no variable other than λ). ..2

$$\mathbf{Answer}: g_0(\lambda) = F(\mathbf{w}^0 + \lambda [1 \ 1]^T) - F(\mathbf{w}^0)$$

$$g_0(\lambda) = F([\lambda \ \lambda]^T) - F([0 \ 0]^T)$$

$$g_0(\lambda) = (1 - 10\lambda)^2 + (1 - 11\lambda)^2 + (-1 - 6\lambda)^2 + (-1 - 9\lambda)^2 + (-1 - 11\lambda)^2 + 2\lambda^2 - 5$$

$$g_0(\lambda) = 461\lambda^2 + 10\lambda$$

Rubric:

- Correctly setting up $g_0(\lambda) = F(\mathbf{w}^0 + \lambda \mathbf{e}^0) - F(\mathbf{w}^0)$: **0.5 mark**.
- Correct substitution $\mathbf{w}^0 = 0$ and $\mathbf{e}^0 = [1 \ 1]^T$, leading to $F([\lambda \ \lambda]^T)$: **0.5 mark**.

- Correct expanded expression for $g_0(\lambda)$ (second-last step), even if not simplified: **0.5 mark**.
- Correct final simplified polynomial in λ : **0.5 mark**.

Notes:

- The expanded (unsimplified) expression is considered a valid final answer, since simplification is required in the subsequent part.
- Omission of either the regularization term ($2\lambda^2$) or the constant offset (-5) results in a deduction of **0.5 mark**.
- An incorrect final polynomial incurs an additional deduction of **0.5 mark**, with the total awarded marks capped at **1 mark** in such cases.
- Merely stating the origin of the terms (e.g., explaining where $2\lambda^2$ or -5 comes from) without incorporating them correctly into $g_0(\lambda)$ receives **0.5 mark**.
- Missing intermediate steps may be penalized under the “missing step” criterion.

(c) Find optimal λ^* for g_0 .

..1

Answer:

$$\begin{aligned}\lambda^* &= \min_{\lambda} g_0(\lambda) \\ g_0(\lambda)' &= 922\lambda + 10 = 0 \\ g_0(\lambda)'' &= 922 > 0 \\ \lambda^* &= -5/461\end{aligned}$$

Rubric:

- Correctly identifying the optimality condition ($g_0'(\lambda) = 0$ or equivalent minimization statement): **0.5 mark**.
- Correct numerical value of λ^* (only $-\frac{5}{461}$ or equivalently $-\frac{10}{922}$ accepted): **0.5 mark**.

Notes:

- No grace marks are awarded in this part.
- Answers with incorrect sign, incorrect simplification, or arithmetic errors receive **0 mark** for the numerical component.
- Leaving the answer in the form of an unsolved equation (e.g., $922\lambda + 10 = 0$) is treated as an incomplete answer and receives **0 mark** for the final value.
- Any numerical value other than $-\frac{5}{461}$ or $-\frac{10}{922}$ is considered incorrect.
- Again, no steps correct answer, is 0.5 mark max.

(d) In the general case, suppose we choose $\mathbf{e}^t$ as the all ones vector for all iterations. Is the algorithm guaranteed to converge to the optimal with this choice of $\mathbf{e}^t$? Justify. [No marks without proper justification.]

..2

Answer: The algorithm may not converge in the general case because the objective value may be constant along that direction. Here is a case. For $d = 2$ it is easy to construct a training dataset with four instances: two (1,0,1)s and two (0,1,-1)s for which the objective becomes

$$\begin{aligned}F(w_1, w_2) &= (2(1 - w_1)^2 + 2(-1 - w_2)^2) + (w_1^2 + w_2^2). \\ g_0(\lambda) &= 2(1 - \lambda)^2 + 2(-1 - \lambda)^2 + 4\lambda^2.\end{aligned}$$

The minimizer for this is $\lambda = 0$, that is $\mathbf{w}$ is unchanged but the gradient is not zero at that point.

Rubric:

- Correct statement of non-guaranteed convergence (e.g., answering “No”): **1 mark**.

- Clear justification using a valid argument, explanation, or counterexample (not necessarily the one shown above): **1 mark**.
- Statement without justification: **0.5 mark at most**.
- Fully justified argument (statement + reasoning): **2 marks (full)**.
- Any response that does not address convergence or uses irrelevant reasoning: **0 mark**.

Notes:

- Since, there was no unique correct justification; any logically sound argument demonstrating lack of guaranteed convergence was acceptable.
- Answers without meaningful justification do not receive full credit.

6. Suppose you need to train a logistic regression classifier under the constraint that the difference between the largest and smallest parameter value is bounded, that is $\max_j w_j - \min_j w_j \leq \gamma$. Show how to use the proximal gradient method to obtain the updates for $\mathbf{w}$.

..4

Answer:

We solve

$$\min_{\mathbf{w}} f(\mathbf{w}) \quad \text{s.t.} \quad \max_j w_j - \min_j w_j \leq \gamma.$$

Using proximal gradient, let

$$\tilde{\mathbf{w}} = \mathbf{w}^k - \alpha \nabla f(\mathbf{w}^k).$$

Then

$$\mathbf{w}^{k+1} = \text{prox}_{\alpha I_S}(\tilde{\mathbf{w}})$$

where

$$S = \{\mathbf{w} : \max_j w_j - \min_j w_j \leq \gamma\}.$$

The constraint is equivalent to:

$$\exists z \in \mathbb{R} \quad \text{s.t.} \quad z \leq w_j \leq z + \gamma \quad \forall j.$$

Thus the projection becomes

$$F_j(z) = \min_{w_j} (w_j - \tilde{w}_j)^2 \quad \text{s.t.} \quad z \leq w_j \leq z + \gamma, \quad \forall j. \quad (1)$$

For fixed z , the solution is coordinate-wise clipping:

$$w_j(z) = \min\{\max\{\tilde{w}_j, z\}, z + \gamma\}.$$

For each j , we use binary search to find the z_j for which $F_j(z)$ is minimized.

Rubric: Full marks if correctly solved the proximal step.

Total: 28
